Worksheet 20 (Cryptography 4): Double Transposition

Group Names: _____

Double Transposition

1. Use the first keyword `RIGHTS` and the second keyword `UNITE`, encrypt the phrase

   E S T A B|L I S H J|U S T I C|E

R	I	G	H	T	S
E	S	T	A	B	L
I	S	H	J	U	S
T	I	C	E		

→

G	H	I	R	S	T
T	A	S	E	L	B
H	J	S	I	S	U
C	E	I	T		

→

U	N	I	T	E
T	H	C	A	J
E	S	S	I	E
I	T	L	S	B
U				

→

E	I	N	T	U
J	C	H	A	T
E	S	S	I	E
B	L	T	S	I
				U

JEBCS LHSTA ISTEI U

Check 16 chara ✓

2. Assuming the ciphertext was encrypted using double transposition with the first keyword `RIGHTS` and the second keyword `UNITE`, decrypt the ciphertext

   Y E S E L|I N F O E|E B S S S B|L I T R R|C U H T E G 27 chara

27/5 = 5 R 2

cols ↓

E	I	N*	T	U*
Y	I	E	L	C
E	N	B	I	U
S	F	S	T	H
E	O	S	R	T
L	E	S	R	E
		B		G

→ ↔

U*	N*	I	T	E
C	E	I	L	Y
U	B	N	I	E
H	S	F	T	S
T	S	O	R	E
E	S	E	R	L
G	B			

→ Rows → cols

27/6 = 4 R 3

G*	H	I*	R*	S	T
C	U	E	S	E	R
E	B	H	T	E	L
I	N	S	S	S	G
L	I	F	O	E	B
Y		T	R		

→

R*	I*	G*	H	T	S
S	E	C	U	R	E
T	H	E	B	L	E
S	S	I	N	G	S
O	F	L	I	B	E
R	T	Y			

Secure the blessings of liberty

3. What are some advantages of this cipher?

It's easy to remember the keywords
surprisingly secure

4. What are some disadvantages of this cipher?

very, very succeptible to transcription errors.

Optional Challenge:

The following ciphertext was encoded using Double Transposition, with first keyword `RIGHTS` and second keyword `ARTICLE`. What was the plaintext?

·ETRNE ·LAAOB ·BRATN EUEES SCROT ·RHSPR ·OIGNO ·HODMA ·TOOTO
·EOUPN ·ENEYN ·RITDE ·PFIAN ·NTDTG ·HTTNS ·RRLIT ·NHHEE ·TEITS
·DEICT ·CSELA ·IOUHE IY ← 107 char

This takes a lot of very careful work.

Start

A C E I L R T

A R T I C L E

$107/7 = 15 R 2$

The enumeration in the constitution of certain rights shall not be construed to deny or disparage others retained by the people.

then

G H I R S T

R I G H T S

$107/6 = 17 R 6$